

Mike Silva

📍 Paris, France

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Research interests: complex systems · quantum information · stochastic processes · machine learning for physics · agent-based and mean-field models · quantitative finance.

Education

Université Paris-Saclay 2026 – 2027 (expected)
MSc Physics of Complex Systems Orsay, France

Coursework: stochastic processes, statistical field theory, disordered systems, random matrices, machine learning.

Sorbonne Université 2026 – 2027 (expected)
MSc Quantum Information Paris, France

Coursework: quantum algorithms, quantum information theory, quantum optics, quantum metrology, ultracold atoms.

Université Paris Dauphine-PSL 2025 – 2027
MSc Economics & Finance, first year Paris, France

Coursework: stochastic calculus, time-series econometrics, derivatives pricing, risk management, computational finance.

Université Paris Cité 2025 – 2026
MSc Fundamental Physics, first year, mention Bien Paris, France

Sorbonne Université 2021 – 2025
MSc Mathematics, first year; BSc Physics and Mathematics (double major) Paris, France

Coursework: probability, stochastic control, jump processes, functional analysis, PDEs, numerical methods, statistical physics.

Université Paris 1 Panthéon-Sorbonne 2022 – 2025
BA Economics and Political Science Paris, France

Coursework: econometric methods, macro- and microeconomics, monetary economics, political economy.

Research Experience

IJCLab – Irène Joliot-Curie Laboratory (CNRS / Université Paris Cité) Jun 2026 – Jul 2026
Research Intern, supervised by Chloé Hebborn and Patrick McGlynn Orsay, France

Emulation of nuclear reaction models for uncertainty quantification on optical potentials. Thesis: *Computational Emulation of Nuclear Dynamics*, graded 18/20 (report and code).

- Developed a coupled-channel scattering solver based on the Direct Boundary Matching Method (DBMM), validated against an independent R-matrix calculation
- Built two emulators (reduced basis method; learning reduced-order model, first coupled-channel extension) matching the solver within 5×10^{-3} (median) on four systems with up to 8 coupled channels and 10 parameters
- Benchmarked both against the solver: speed-ups grow with system size, and the learning model gains 1.3–1.9× more

CEREMADE, Université Paris Dauphine-PSL Sep 2024 – Jan 2025
Visiting Research Student, supervised by Pierre Cardaliaguet Paris, France

Mean-field games applied to the propagation of monetary shocks in a general-equilibrium economy with sticky prices.

- Studied differential games and mean-field models of strategic interactions between heterogeneous agents
- Analyzed the coupled Hamilton–Jacobi–Bellman and Fokker–Planck equations arising in dynamic economic models
- Applied mathematical and computational tools to monetary policy and collective decision-making

Sciences Po Jun 2023 – Aug 2023
Research Assistant, supervised by Julia Cagé Paris, France

Built a large-scale dataset of French television broadcasts for a study of media ownership and content bias.

Research Projects

Urban morphogenesis: diffusion-limited aggregation and Schelling’s model Jan 2026
Université Paris Cité, supervised by Raphael Raynaud Paris, France

Agent-based model coupling Schelling segregation with diffusion-limited aggregation: emergent morphologies and fractal dimension; extension with reinforcement learning under way.

Self-gravitating N-body systems Feb 2024
Sorbonne Université, supervised by Lionel Foret Paris, France

Direct-integration simulation of gravitational N-body dynamics and long-range collective effects.

Skills

Programming	Python (NumPy, SciPy, Numba, JAX, PyTorch), R; Git, pytest, CI, LaTeX
Numerics & ML	PDE solvers, reduced-order models, Monte Carlo, neural networks, reinforcement learning
Quantum	quantum algorithms, quantum information theory, spin glasses, random matrices
Complex systems	agent-based models, mean-field games, stochastic processes, fractal growth
Economics	stochastic calculus, derivatives pricing, econometrics, general equilibrium
Languages	French (native), English (fluent), Portuguese (basic), German (basic)